Amit Amit

Data Scientist | Computer Vision and Applied AI

Gotha, Germany | linkedin.com/in/amit-tech
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Professional summary

Data Scientist at Sedeco Imaging building camera-based detection and robotic sorting workflows. M.Sc. Data Science graduate with research experience in computer vision, scientific machine learning and biomedical data, plus prior Linux, Python/Bash and SQL experience.

Technical skills

Machine learning: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, deep learning, anomaly detection, segmentation, physics-informed neural networks (PINNs).
Data and software: Python, SQL, NumPy, Pandas, Git, Docker, Linux, Django, TypeScript/Next.js.
Computer vision and edge: OpenMV, Raspberry Pi, Luckfox, Hikrobot camera integration.

Professional experience

Data Scientist | Sedeco Imaging

Gotha, Germany | October 2025 - Present

- Lead the technical design of bottle and can detection workflows and robotic separation using an FR5 arm.
- Develop camera-based vision components and deployment solutions for compact platforms including Luckfox, OpenMV and Raspberry Pi, alongside Hikrobot systems.

Student Research Assistant | PLRI, TU Braunschweig

Braunschweig, Germany | November 2022 - September 2025

- Analyzed multi-omics data using network-based methods for NetfLID; supported gene-gene interaction modeling for the Epistasis Disease Atlas.
- Built support vector machine (SVM) models and visualizations for biomedical researchers; supported Linux infrastructure and contributed to a high-performance computing (HPC) cluster.

Student Research Assistant | German Aerospace Center DLR

Braunschweig, Germany | November 2024 - June 2025

- Developed convolutional neural network (CNN) models of structural wear using inspection images and environmental data for aircraft component lifecycle assessment.
- Investigated data provenance and blockchain approaches for lifecycle assessment; coauthored a conference contribution listed by DLR.

Student Assistant | iBMB, TU Braunschweig

Braunschweig, Germany | March 2023 - November 2024

- Prepared experimental data and developed predictive models, including Random Forests, for materials and fire-safety research.
- Built a Django application for internal experimental-data management and researcher access.

Senior Systems Engineer | Infosys

Bengaluru, India | March 2019 - November 2021

- Supported Linux and VMware infrastructure; used SQL for analysis and reporting and Python/Bash scripts for recurring operational tasks.

Selected projects

Industrial visual inspection | Personal research

2026 | Python, PyTorch, CUDA | Case study and recorded examples

- Built a reproducible command-line interface (CLI) for data preparation, normal-only calibration, prediction, evaluation and benchmarking; implemented a NumPy baseline and GPU PatchCore variant.
- Compared preprocessing on a PCB2 development split: cropped 512 px input detected 43/50 defects with 1/50 false alarms, versus 36/50 and 3/50 for full-frame 256 px input.
- Evaluated transfer to a second component dataset and documented poor performance. Results are from development experiments; independent validation is still pending.

Physics-informed neural networks | Master's thesis

2025 | TensorFlow, Hydra, scientific computing | Repository

- Investigated PINN training for Helmholtz and Allen-Cahn equations using existing frameworks including TensorDiffEq and pinns-tf2.
- Implemented experiments exploring collocation, adaptive weighting and optimization, with saved solution, residual and error visualizations.

Landslide segmentation | Collaborative academic project

2023 | UNet, Swin-based UNet, SegFormer | Team report

- Collaborated with Reiko Lettmoden on multispectral segmentation, comparing architectures, input bands, augmentation and imbalance-aware losses.
- The team report records F1 0.756 for augmented UNet with Lovasz loss versus 0.721 baseline, using a local split of a training subset, not the official benchmark test set.

Additional applied AI and research projects

2022 - 2025 | Portfolio archive

- Speech enhancement: CPC representations and HiFi-GAN in PyTorch; NISQA, PESQ and DNSMOS evaluation against an EffCRN baseline.
- Character AI application: Next.js/TypeScript, Express, MongoDB and OpenRouter integration. Earlier work includes equivariant leakage localization and water forecasting.

Education

M.Sc. Data Science | Technische Universität Braunschweig
October 2021 - June 2025

B.Tech. Electronics and Communication Engineering
Bharati Vidyapeeth's College of Engineering, New Delhi | August 2014 - June 2018

Conference contribution

Janko Kreikemeier, Amit Amit and Steffen Opitz. *Towards a Circular Future with Structural Life 2.0 and Blockchain-Approach.* World Symposium on Materials Sciences and Engineering, Singapore, December 2025. Conference speech; DLR bibliographic record.

Languages

English: C1 | German: A2